

ADVANCED REAL ANALYSIS

II. ABSTRACT INTEGRATION THEORY

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ABSTRACT

These are some notes I made while reading through Folland's book [Fol99]. Real Analysis is a very difficult subject, especially at the graduate level. As such, I figured I'd make them public so other learners have more resources to study.

Hopefully, graduate students and those studying for PhD qualifying examinations in Analysis will find them useful.

Comments and suggestions to improve this manuscript are most welcome:

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Please also email me if you discover any errors.

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*To Ayla, for being a great hiking partner in the few days we were together.
Hopefully you enjoyed the waterfalls!*



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1. REVIEW ON PREIMAGES

Before defining a measurable function, a review of preimages.

Consider a map $f : X \rightarrow Y$. It is useful to be able to keep track of the points in the domain that end up in a specific subset of the target space (the range) in a set. Formally, we call these sets *preimages*¹.

Definition: Preimages

For a map $f : X \rightarrow Y$, consider a subset $E \subset Y$. The *preimage* of E under f is the set

$$f^{-1}(E) = \{x \in X : f(x) \in E\}.$$

That is to say the preimage consists of the points in the function's domain that get mapped into the target set E . The next theorem will be useful. Its proof is easy, and follows directly from the definition. I shall prove (1), and leave the rest as an exercise. Read my proof, then attempt (2) and (3)—You shouldn't have much trouble doing this, and you may check your answer by referring to 2.33 in [Axl20].

Proposition 1.

Consider a map $f : X \rightarrow Y$, and let $\mathcal{A}$ be a set of subsets of Y . The preimage of this function satisfies the following properties:

- (1) $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$
- (2) $f^{-1}(\cup_{A \in \mathcal{A}} A) = \cup_{A \in \mathcal{A}} f^{-1}(A)$
- (3) $f^{-1}(\cap_{A \in \mathcal{A}} A) = \cap_{A \in \mathcal{A}} f^{-1}(A)$.

Proof:

(1): By definition, $x \in f^{-1}(Y \setminus A)$ if and only if $f(x) \in Y \setminus A$. (and $x \in X$). This is true if and only if $f(x) \notin A$, which is true if and only if $x \notin f^{-1}(A)$. In particular, we have $x \in f^{-1}(Y \setminus A)$ if and only if $x \in X \setminus f^{-1}(A)$, so we have double inclusion between the two sets and the claim follows.

(2), (3): Warm up exercise. ■

¹Preimages are an extremely useful concept in Mathematics, and can be used to provide a clean condition of continuity on metric/topological spaces, a fact we shall use later.

We also have a result on the preimage of a composition:

Proposition 2.

Consider maps $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Then for any set $A \in Z$,

$$(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A)).$$

Proof: Let $A \subset Z$ be given. By definition, $x \in (g \circ f)^{-1}(A)$ iff $(g \circ f)(x) \in A$, the case iff $f(x) \in g^{-1}(A)$, which holds iff $x \in f^{-1}(g^{-1}(A))$. ■

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